

Distributive & Expanding — Practice

Section A is interactive. Sections B and C are written practice — check them against the separate answer key.

Objective (1 mark each)

Q1 • MCQ**[1]**

$a(b + c)$ equals:

(a) $ab + c$ (b) $ab + ac$ (c) $a + bc$ (d) abc

Q2 • MCQ**[1]**

$(a + m)(b + n)$ expands to:

(a) $ab + mn$ (b) $ab + an + mb + mn$ (c) $ab + an + mn$ (d) $am + bn$

Q3 • FILL IN THE BLANK**[1]**

$5(x + 3) = 5x + \underline{\hspace{2cm}}$.

Q4 • MCQ**[1]**

$-3p(-5p + 2q)$ equals:

(a) $15p^2 - 6pq$ (b) $-15p^2 + 6pq$ (c) $-3p + 5p - 2q$ (d) $15p^2 + 6pq$

Q5 • TRUE / FALSE**[1]**

a^2b and ab^2 are like terms.

True False

Q6 · ASSERTION-REASON**[1]****Assertion (A):** $23 \times 27 + 23 = 23 \times 28$.**Reason (R):** The distributive property gives $a(b + c) = ab + ac$.

- (a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false
(d) A false, R true

Short answer (2 marks each)

Q7**[2]**Expand: (i) $(3 + u)(v - 3)$ (ii) $(5 + z)(y + 9)$.

Q8**[2]**Expand $(10a + b)(10c + d)$.

Q9**[2]**Simplify $3a^2b + 2a^2b + 4ab^2$.

Long answer (3 marks each)

Q10**[3]**Expand and simplify $(3 - x)(x - 6)$.

Q11

[3]

Expand $(a + ab - 3b^2)(4 + b)$.

Q12

[3]

Expand $(4y + 7)(y + 11z - 3)$.

Total: 21 marks · Check your work against the [answer key](#).